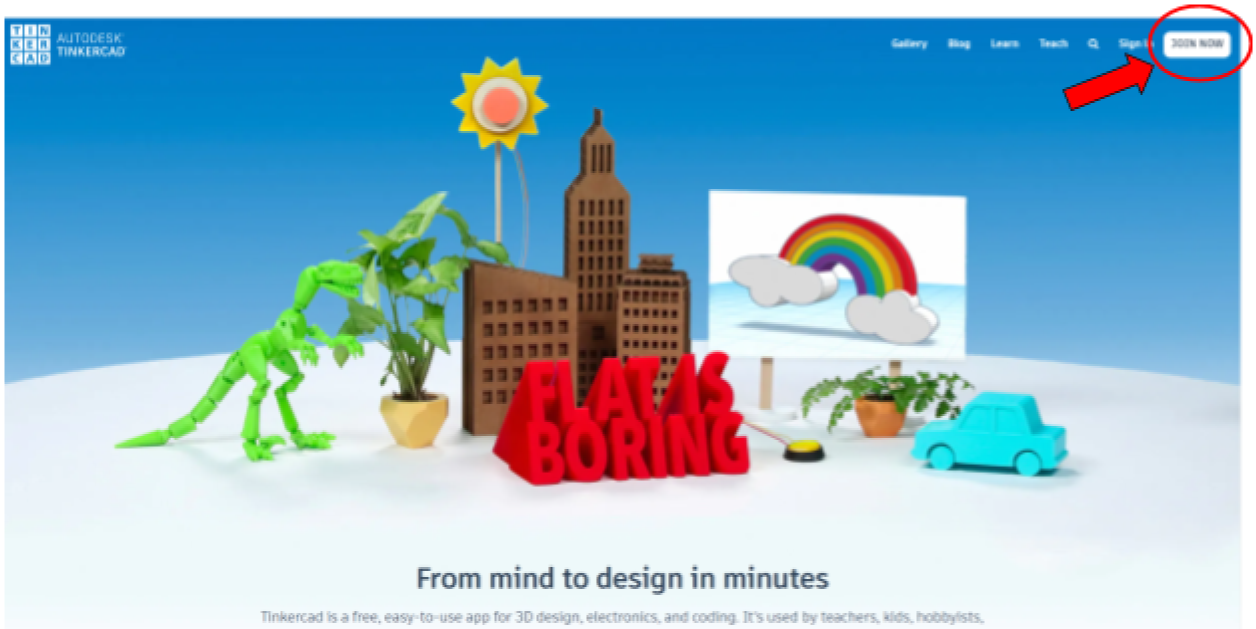
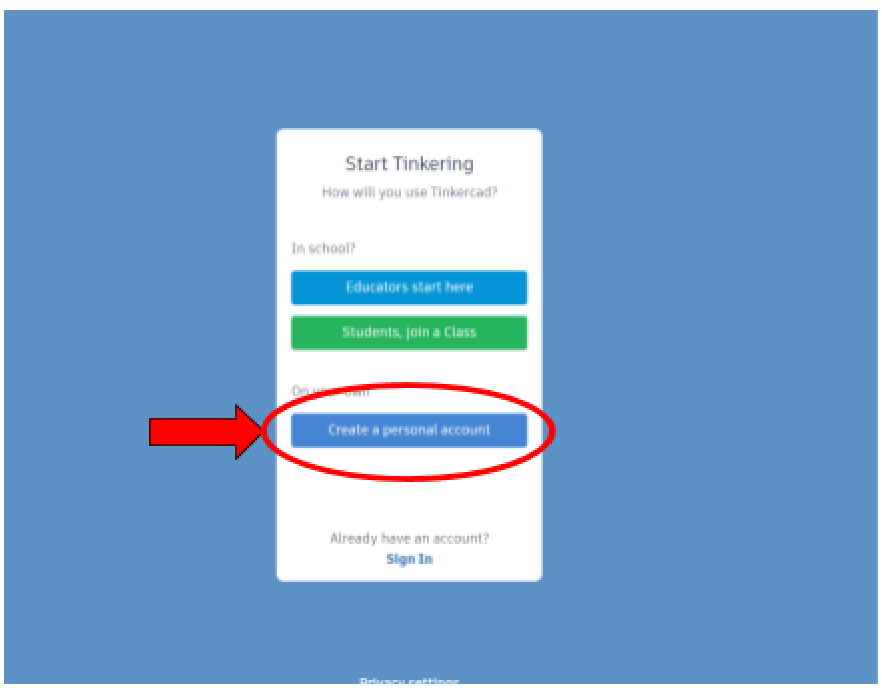
I. Getting Started with Tinkercad

1. Go to tinkercad.com and hit the JOIN NOW button on the top right corner of the website.

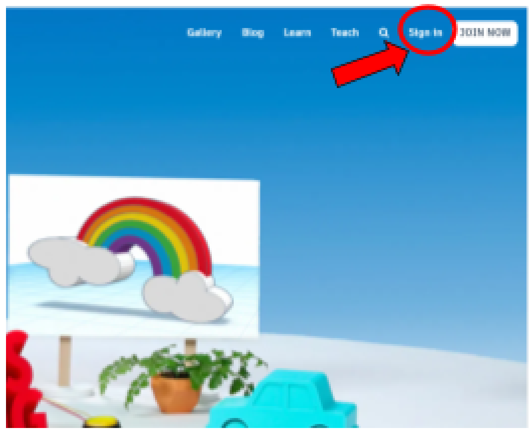


2. Click on "Create a personal account" and proceed with either a Google account or sign up with a different email.

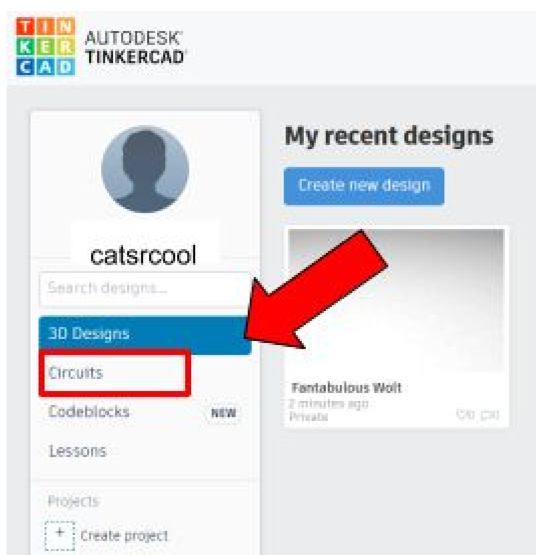


3. Follow the steps and finish creating your Tinkercad account.

4. In the future, to login, hit the Sign In button next to the JOIN NOW button on the website home page and click on the appropriate button (the Email or Username button or the Sign in with Google button).



5. You should now see your designs dashboard. If not, click on your user portrait in the top right corner and select Designs.
6. On the left side of the screen, you should see a list of design types: 3D Designs, Circuits, Codeblocks, and Lessons. Click on Circuits, and hit Create new Circuit.



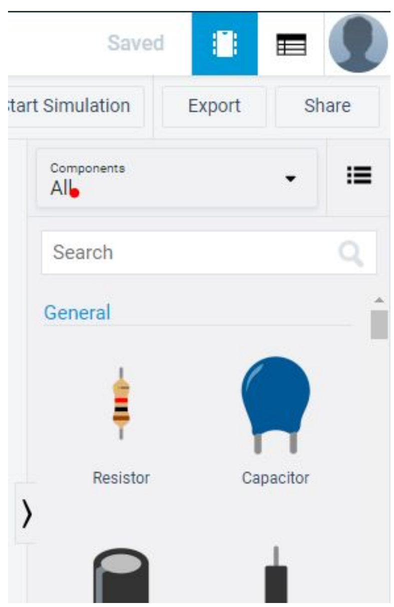
7. You're now ready to start making your circuits!

II. Tinkercad Tutorials: Your First Circuits

A. Things to Know

1. Along the top, you should see the buttons: Rotate, Delete, Undo, Redo, Annotation, View/Hide, Code, Start Simulation, Export, and Share. This is your toolbar.

2. On the right side of your screen, you should see a drop down menu below your toolbar. Clicking on it should reveal the choices of Basic and All Components, as well as some starter circuits. Select All Components, and you will see a list of all of the components you can use for your circuits, along with what they look like. This ranges from basic components like resistors and capacitors to microcontrollers like the Arduino Uno.

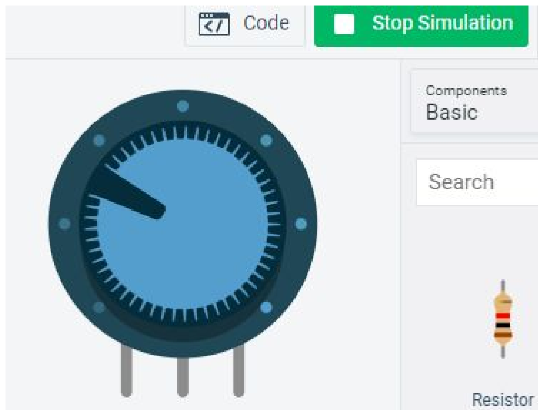


3. To place a component, simply click on the desired component and click again in your workspace to place it down. You should see the image underneath your cursor to help you guide where you place the components.

4. If you want to rotate the component before placing it down, press “R” on your keyboard until you get your desired orientation. Press “F” to flip the component. If you’ve already placed the component and want to rotate or flip it, simply click on the component and press the same keys as above to do so, or use the Rotate button on your toolbar.
5. You can name the component and set other variables, like its value, in the window that pops up. For example, when you place down a resistor, you’re able to set the name and value of the resistor, like R3 and 100kΩ. Click anywhere in your workspace outside of the pop up or choose another component to place to make the pop up disappear. If you ever want to change one of its variables again, simply click on the component and the pop up will reappear.

6. If you change your mind about placing the component, press escape before you place it down. You should see the image following your cursor disappear.
7. To move components around, simply drag the components around with your cursor. Click and drag anywhere in your workspace that’s not a component to pan around the workspace. You can zoom in and out of your workspace by scrolling.

8. Some components are interactive, and you are able to manipulate the object and sometimes its properties by clicking and dragging certain areas on the component. For example, under the simulation mode, the potentiometer has a knob that you can turn in order to change its resistance. Other components like the breadboard have holes that you can click on to create wires to elsewhere on the board.



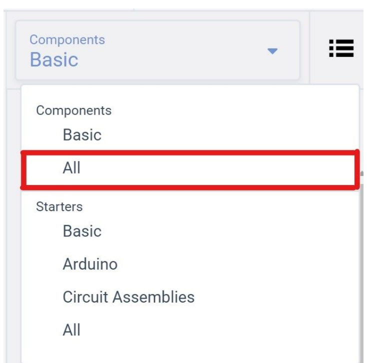
9. To delete a component, click on it and press “Delete” on your keyboard, or click the delete button (trash can symbol) on your toolbar.
10. If you mess up and want to undo something, simply use the same keyboard shortcuts as you would use to normally undo something on your computer (e.g. Ctrl Z for Windows). Or, click on the undo button on the toolbar. The same rules apply for redos as well.
11. Once you feel comfortable with the interface, feel free to get started with our mini-project tutorials! Tinkercad also has its own basic tutorials available on its website that you can explore as well.

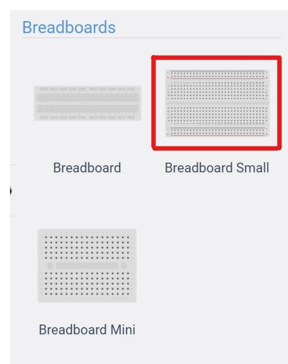
B. Blink

You're now ready to start making your first circuit! In this tutorial, we will be guiding you through making the basic Blink circuit that triggers an LED to turn on and off repeatedly.

Circuitry

1. Create a new circuit.
2. Change your components list from Basic to All and begin by placing down a small breadboard (under the "Breadboards" components list). A breadboard is a basic component that enables you to connect various other components such as resistors and LEDs in a neat and organized manner. You will be using this for almost all of the circuits you make in Tinkercad.



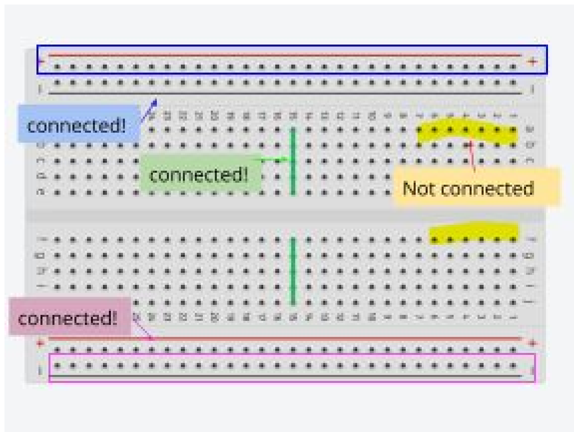


3. A brief introduction to breadboards:
    a. On the breadboard, you should see numbered columns with lettered rows (or numbered rows with lettered columns, if your breadboard is vertical). Each number's letters are connected to each other, but not across the middle divider. For example, column 1's ABCDE are connected together and FGHIJ are connected, but those 2 groups are not connected to each other.
    b. Furthermore, the 2 rows of holes on either side of the breadboard, called rails, labelled + and - are connected across the entire row (all holes on the + rail are connected, all holes on the - rail are connected,

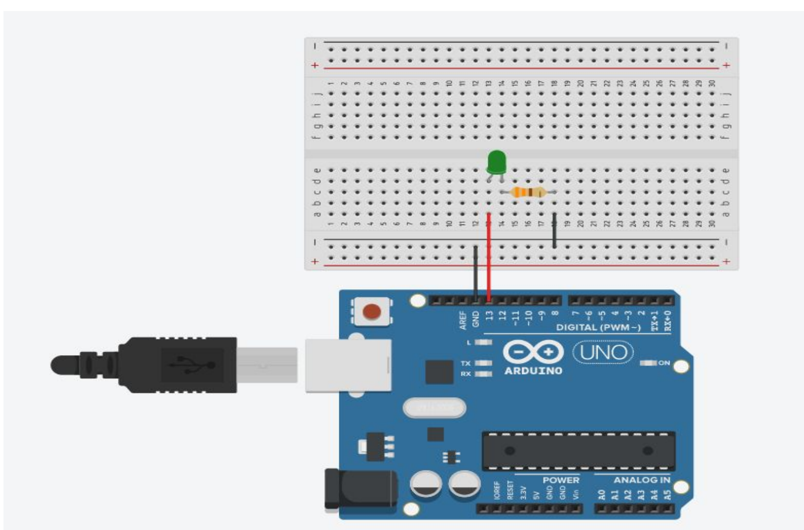
but those rails are not connected each other or with the rails on the other side of the breadboard).

c. When you hover over a hole on the breadboard in Tinkercad, you'll be able to see which other holes it's inherently connected to on the breadboard.

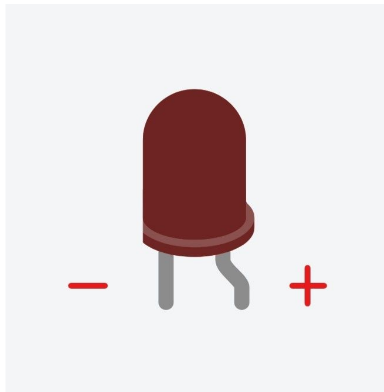
d. Breadboard diagram:



4. Place down an Arduino Uno component and build the circuit on the breadboard as shown. Modify the resistor value to be 330 Ω. You can add wires to make connections by clicking on one of the holes and then clicking on the hole you want to connect it to. Press escape to cancel adding a wire. You can change the color of the wire by clicking on it.



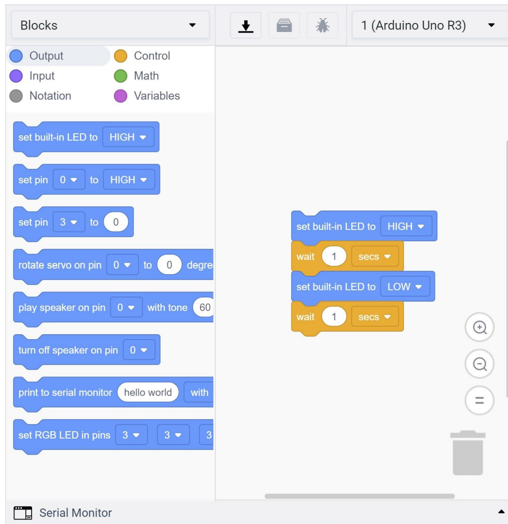
a. Be sure to pay attention to the orientation of the LED! LEDs are polarized, meaning they will only work if you plug them in in the right direction. Make sure the cathode, or the negative side of the LED (the straight leg), is connected to the resistor, and the anode, or the positive side (the bent/crooked leg), is connected to the Arduino.



5. You don't need to make your circuit look exactly like what's shown. So long as all of the electrical connections are where they need to go, your circuit should behave the same way.
6. You've now got all the hardware you need to make the LED blink! Now all that's left to do is to program the Arduino to do so.

Coding

1. We'll now give a brief overview of how coding in Tinkercad works. In order to be able to code, you need to have a component in your circuit that is able to be coded. In this case, it is your Arduino Uno that you have placed down. Click the Code button on your toolbar to pull up the coding interface.



   a. On the top right of the interface, you'll see a drop down menu with the name of the component you're programming. If you have more than one programmable component in your circuit, you'll be able to choose which to program. For now, you only have one programmable component, so there should be only 1 option.
   b. On the top left of the interface, you'll see a drop down menu with the options of Block, Block + Text, and Text. Block code enables you to code by dragging and dropping existing blocks of code and connect them to form the logic flow. Text enables you to code in the traditional format by typing. Block + Text lets you code with blocks while seeing the translation in text.
   c. As most of you should have some form of programming experience already, we will be using the Text option.

d. You will see 2 main functions in the code: setup() and loop(). setup() is run once when you first start the simulation, reset the microcontroller, or in real life, when you power on the microcontroller. It is used to run any one time setups, like changing the mode of certain pins to input or output. loop() is run over and over again. Put all code you want the controller to run here.

e. When your code is all done, press the Start Simulation button on your toolbar and your code will be run! The simulation will keep running your code over and over in a loop, until you press Stop Simulation or an error is encountered.

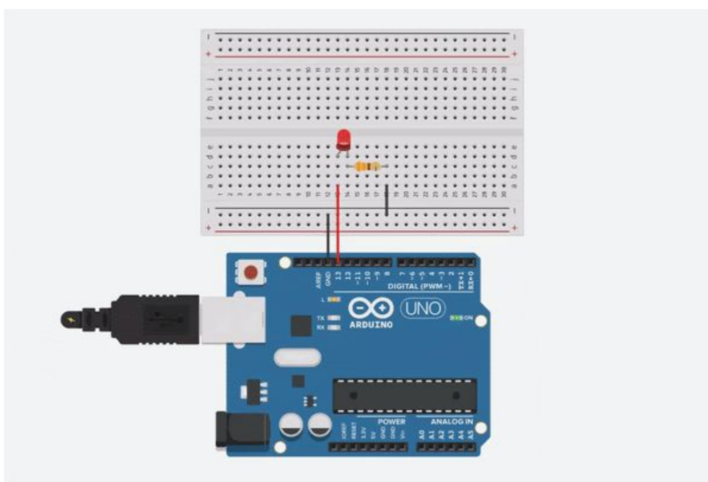
f. On the bottom of the interface, there is a button labelled Serial Monitor. This enables you to view any outputs that the Arduino prints or displays during simulation if you choose to code it to do so. You must include the line Serial.begin(9600); or Serial.begin(38400); in setup() to be able to use it. After doing so you can call functions like Serial.println() to output things to the Serial Monitor.

2. Once you are comfortable with the coding interface, set up the pin your circuit is connected to as an output in setup(). Then in loop(), you can use digitalWrite(pinNumber, value), with value being *high* or *low*, to turn the LED on and off. Make sure to use a delay(delay_milliseconds) in between turning it on and off.

3. Your code should look something like this:

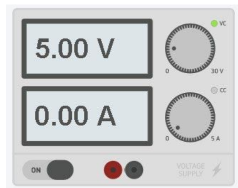
```
void setup()
{
  pinMode(13, OUTPUT);
}

void loop()
{
  digitalWrite(13, HIGH);
  delay(1000); // Wait for 1000 millisecond(s)
  digitalWrite(13, LOW);
  delay(1000); // Wait for 1000 millisecond(s)
}
```

4. Click the Start Simulation button on the toolbar to run your code. You should see the LED turn on and off with a 1 second delay between the 2 states.
5. Feel free to play around with the wait time to make the LED blink at a different rate, or experiment with different code blocks to make the LED turn on at different times. You can also add to the circuit if you would like to experiment with that as well.
6. Congratulations! You have now built, coded, and simulated your first circuit in Tinkercad!



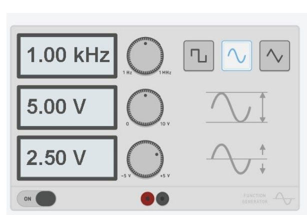
C. Useful Tools

1. There are some useful tools under the Instruments section under all components that model the tools we use for circuits in the lab. Note that they only display values when you start the simulation.
2. First up is the power supply. You can set it to supply specific voltages with current limits. In this class, always set your current limits to .100A unless otherwise specified. Connect the positive (red) to where you want to supply the voltage to and the negative (black) to gnd.



3. Next, we have the multimeter. You can measure voltage, current, and resistance with the multimeter. Connect the probes/pins on the multimeter to your circuit in the direction of the current flow to measure positive values.

4. Third, we have the waveform generator. This allows you to generate various voltage signals, such as sin and square waves. You can change the frequency, amplitude, DC offset (constant offset in the voltage), and the shape of the waveform. Once again, connect positive and negative to the input of your circuit and gnd, respectively.



5. Finally, there is the oscilloscope. The oscilloscope is able to measure and display voltage waveforms in time. The voltage scale will be automatically set depending on what you measure, and you can adjust the time scale

manually. Connect the positive terminal to what you want to measure and the negative terminal to gnd.